

JAVASCRIPT ESSENTIALS:

Javascript is a popular scripting language that allows web developers to add interactive features to websites containing HTML and CSS. you can add interactivity like validation, onclick actions, animation etc through JS.

- **Variables:** Variables are containers that are used to store data inside them. There are three ways to declare variable in JS:
 - a) **Var** : its function scoped
 - b) **let and const** : they are block scoped, including better control over variable visibility within specific code blocks
- **Data types:** It tells the type of value a variable can hold. Ex: string, number, boolean, null, undefined, and object (for more complex data like arrays or objects)

- **Function:** A function represents a block of code designed to perform a specific task.
- **Loops:** Loops allow you to run a code block multiple times as long as a condition is true. Common loops in JS are for, while, do...while.
- **Javascript** is a interpreted language, where the code is executed directly in the browser without prior compilation.

Ex: Javascript: // Hello world program
 console.log("Hello, World!");

// Variable and data type

let a = 5; // number type

// Control flow statement

```
if (a ≥ 5) {  
    console.log("I am saransh");  
}
```

```
else {  
    console.log("I am Pahadi");  
}
```

// function

```
function greet(name) {  
    console.log("Hello, " + name + "!");  
}
```

// Calling the function

greet("Bob");

Javascript is basically executed on client side. ~~was~~

- How to open google chrome console?
ctrl + shift + i or right click and select inspect
then click the console tab.
- There are two main ways to integrate JS into HTML: internal and externally

a) Internal JS: Internal JS means embedding the JS code directly within an HTML document. The script is inserted between `<script>` tags. These tags can be placed in head section, typically used for scripts that need to be loaded before the page content is rendered or inside `<body>` section, where script can be utilized to interact with elements as they are loaded on web page.

b) External JS: External JS involves creating and storing JS code in a separate file ending with a .js file extension. The external JS can be stored or hosted on the same web server as the HTML document.

We can load the JS code by using src =js attribute inside the HTML document to load the JS file ~~and~~ while running. But it keeps both JS and HTML code separately.

NOTE: when pen-testing a web application, it is important to check whether the website uses internal or external JS which can be seen using view page source.

- One of main objectives of JS is to provide dialogue boxes for interaction with users and dynamically updates content on web pages.

JS provides built in function like alert, prompt and confirm to facilitate this interaction.

- a) Alert: The alert function displays a message in a dialogue box with an "ok" button, typically used to convey information or warning to users.

Ex: `alert("Saranish")` → run it on chrome console.

- b) Prompt: The prompt function displays a dialogue box that asks the user for input. It returns the entered value when the user clicks "ok" ~~and~~ or null if the user clicks "cancel".

Ex: `name = prompt("what is your name?");`

~~and~~ `alert("Hello" + name);`

↓
Prints the info using alert functions.

↑
takes input and save it in name

- c) Confirm: The confirm function displays a dialogue box with a message and two buttons "ok" and "CANCEL". It returns true if user clicks ok and returns false if user clicks cancel.

Ex: `confirm("Are you sure?");`

- Control flow in JS refers to the order in which statements and codes are executed based on certain conditions. JS provides several control flow structures such as if-else, switch,

for, while, do... while.

- **Minification** : It is the process in ~~pp~~ ~~in~~ which JS files are compressed by removing all unnecessary characters, such as spaces, line breaks, comments and even shortening variable name. Minified files makes code more compact and harder to read for humans.

Similar Obfuscation is often used to make JS harder to understand by adding undesired code, renaming variable and functions and even adding dummy code.



This can be done by online tool called JavaScript Obfuscator.

It can also be deobfuscated using online tools